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(54) Title of the invention : AUTOMATED MILK COLLECTION CENTRE FOR INDIAN VILLAGES

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(57) Abstract :

Abstract These days the world is quicker. In everyday life prerequisites, milk takes up significant spot. So this proposed framework will change Indian rancher"s way of life. Town Level Milk Collection Center In-control is liable for obtaining and taking care of huge volumes of milk from numerous individual milk vendors. We can give a brilliant framework to gather the milk from the people and paying them. The amount alongside the card number will be put away in the cloud. Consistently the bills will be created for each rancher, so that in-control can pay them. Administrator will be given a versatile App utilizing which he can screen all boundaries. Milk dissemination and wellbeing is of high worry as it includes the strength of 90% of our general public. Our venture, milk appropriation observing framework focuses on the virus chain upkeep and milk waste shirking. The framework depends on Internet of Things and information mining. Generally milk is provided in jars with least checking which may bring about milk getting ruined before its utilization, particularly during transportation anytime of time which makes merchants think about milk waste simply after the milk has been ruined totally. The sensor based framework will gauge the pH estimation of milk and decide its quality. The all out number of milk stall in every zones whose information is open any time in the organization. The load cell present in the framework gauges the amount of the milk.

No. of Pages : 12 No. of Claims : 6

Application Details

APPLICATION NUMBER	202041049776
APPLICATION TYPE	ORDINARY APPLICATION
DATE OF FILING	13/11/2020
APPLICANT NAME	1 . Dr C G RAVICHANDRAN 2 . Dr. S. GOPIKRISHNAN 3 . SHEMITHA P A 4 . V.ANUSUYA 5 . Dr. R. SABITHA 6 . Dr.R.INDHUMATHI 7 . Dr. ARUN.P 8 . Dr.A.KINGSLY JABAKUMAR 9 . Dr. S. A. SIVAKUMAR
TITLE OF INVENTION	AUTOMATED MILK COLLECTION CENTRE FOR INDIAN VILLAGES
FIELD OF INVENTION	AGRICULTURE ENGINEERING
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PRIORITY DATE	
REQUEST FOR EXAMINATION DATE	--
PUBLICATION DATE (U/S 11A)	20/11/2020

Application Status

APPLICATION STATUS	Awaiting Request for Examination
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[View Documents](#)

➡ Filed ➡ Published ➡ RQ Filed ➡ Under Examination ➡ Disposed

FORM 1
THE PATENTS ACT, 1970
(39 of 1970)
&
THE PATENTS RULES, 2003
APPLICATION FOR GRANT OF PATENT
[See sections 7,54 & 135 and rule 20(1)]

(FOR OFFICE USE ONLY)

Application No.:
Filing Date:
Amount of Fee Paid:
CBR No.:
Signature:

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3. TITLE OF THE INVENTION: AUTOMATED MILK COLLECTION CENTRE FOR INDIAN VILLAGES

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5. PRIORITY PARTICULARS OF THE APPLICATION(S) FILED IN CONVENTION COUNTRY:

Sr.No.	Country	Application Number	Filing Date	Name of the Applicant	Title of the Invention
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6. PARTICULARS FOR FILING PATENT COOPERATION TREATY (PCT) NATIONAL PHASE APPLICATION:

International Application Number	International Filing Date as Allotted by the Receiving Office
PCT//	

7. PARTICULARS FOR FILING DIVISIONAL APPLICATION

Original (first) Application Number	Date of Filing of Original (first) Application
-------------------------------------	--

8. PARTICULARS FOR FILING PATENT OF ADDITION:

Main Application / Patent Number:	Date of Filing of Main Application
-----------------------------------	------------------------------------

9. DECLARATIONS:

(i) Declaration by the inventor(s)

I/We ,Dr C G RAVICHANDRAN,Dr. S. GOPIKRISHNAN,SHEMITHA P A,V.ANUSUYA,Dr. R. SABITHA,Dr.R.INDHUMATHI,Dr. ARUN.P,Dr.A.KINGSLY JABAKUMAR,Dr. S. A. SIVAKUMAR, is/are the true & first inventor(s) for this invention and declare that the applicant(s) herein is/are my/our assignee or legal representative.

(a) Date: -----

(b) Signature(s) of the inventor(s):

(c) Name(s): Dr C G RAVICHANDRAN,Dr. S. GOPIKRISHNAN,SHEMITHA P A,V.ANUSUYA,Dr. R. SABITHA,Dr.R.INDHUMATHI,Dr. ARUN.P,Dr.A.KINGSLY JABAKUMAR,Dr. S. A. SIVAKUMAR

(ii) Declaration by the applicant(s) in the convention country

I/We, the applicant(s) in the convention country declare that the applicant(s) herein is/are my/our assignee or legal representative.

(a) Date: -----

(b) Signature(s) :

(c) Name(s) of the singnatory: Dr C G RAVICHANDRAN,Dr. S. GOPIKRISHNAN,SHEMITHA P A,VANUSUYA,Dr. R. SABITHA,Dr.R.INDHUMATHI,Dr. ARUN.P,Dr.A.KINGSLY JABAKUMAR,Dr. S. A. SIVAKUMAR

(iii) Declaration by the applicant(s)

- The Complete specification relating to the invention is filed with this application.
- I am/We are, in the possession of the above mentioned invention.
- There is no lawful ground of objection to the grant of the Patent to me/us.

10. FOLLOWING ARE THE ATTACHMENTS WITH THE APPLICATION:

Sr.	Document Description	FileName
-----	----------------------	----------

I/We hereby declare that to the best of my/our knowledge, information and belief the fact and matters stated hering are correct and I/We request that a patent may be granted to me/us for the said invention.

Dated this(Final Payment Date): -----

Signature:

Name: Dr S A Sivakumar

To The Controller of Patents

The Patent office at CHENNAI

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FORM 2
THE PATENTS ACT 1970
(39 of 1970)
&
The Patents Rules, 2003
COMPLETE SPECIFICATION
(See Section 10 and rule 13)

AUTOMATED MILK COLLECTION CENTRE FOR INDIAN VILLAGES

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AUTOMATED MILK COLLECTION CENTRE FOR INDIAN VILLAGES

Abstract

These days the world is quicker. In everyday life prerequisites, milk takes up significant spot. So this proposed framework will change Indian rancher's way of life. Town Level Milk Collection Center In-control is liable for obtaining and taking care of huge volumes of milk from numerous individual milk vendors. We can give a brilliant framework to gather the milk from the people and paying them. The amount alongside the card number will be put away in the cloud. Consistently the bills will be created for each rancher, so that in-control can pay them. Administrator will be given a versatile App utilizing which he can screen all boundaries. Milk dissemination and wellbeing is of high worry as it includes the strength of 90% of our general public. Our venture, milk appropriation observing framework focuses on the virus chain upkeep and milk waste shirking. The framework depends on Internet of Things and information mining. Generally milk is provided in jars with least checking which may bring about milk getting ruined before its utilization, particularly during transportation anytime of time which makes merchants think about milk waste simply after the milk has been ruined totally. The sensor based framework will gauge the pH estimation of milk and decide its quality. The all out number of milk stall in every zones whose information is open any time in the organization. The load cell present in the framework gauges the amount of the milk.

Background of the Invention

The Milk Collection station is a specially designed, integrated unit, which combines the several functions of a milk collection center. It measures the weight and gives the price of the milk brought in by the each villager owning cattle. The equipment is particularly useful for the milk co-operatives / milk collecting center as it can also maintain a summary of milk supplied.

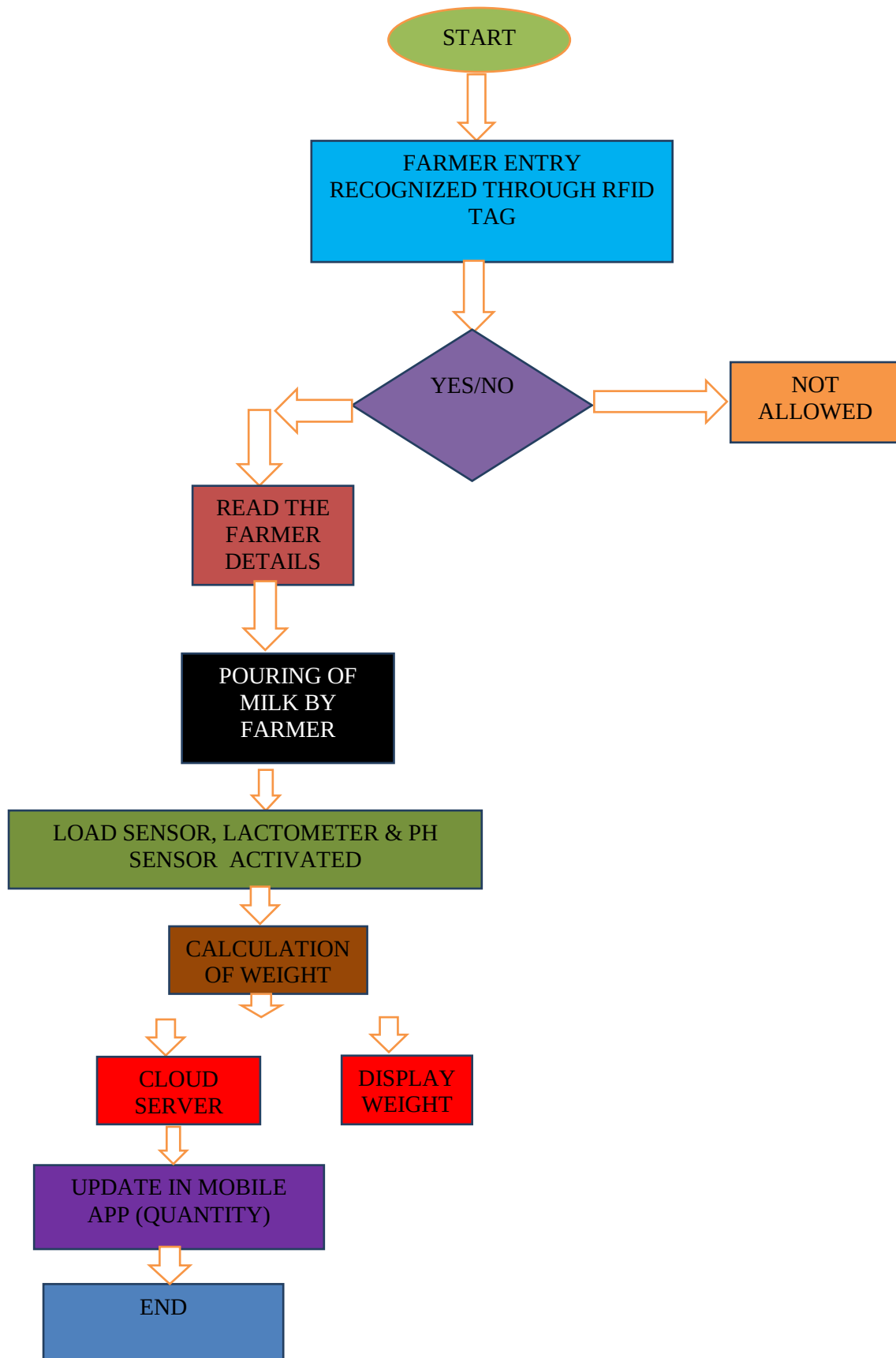


FIGURE 1: FLOW CHART OF THE AUTOMATED MILK COLLECTION CENTRE FOR INDIAN VILLAGES

Milk utilization is one of the most essential requirements for the human, to move milk to satisfy the fundamental human need is a serious testing task. In the previous stages bikes and trucks were utilized to move milk, as the innovation progressed engine vehicles were presented however as the populace expanded dramatically the need to ship milk with absolute exactness got one of the most trying for the milk business. To keep away from decay and save great quality milk different innovative systems were utilized. In customary occasions, milk containers utilized be conveyed from milk vehicles to the clients yet it had no management and frequently milk used to be squandered as sellers came to think about the decay a lot later. New strategies have been investigated to grow more precise and effective methods for distinguishing milk deterioration which incorporates PH and temperature checking. At milk securing stations in provincial regions quality checks must be made mandatory to isolate the milk as indicated by their quality. To acknowledge the milk, fundamental tests are made utilizing pH sensor for finding the corrosiveness or alkalinity of the milk, Lactometer is utilized to realize the water content in the milk, temperature sensor is utilized to get Corrected Lactometer reading(CLR), shading sensor for getting the shade of the milk as it is realized that a large portion of the non-white milk are unfortunate.

Description of the Invention

Nearly with every single Indian town, absence of mindfulness about plans and appropriations in different regions by the administration is the serious issue they face. This has made a significant deterrent on the events when we attempted to refresh advancements and execute a few tasks like bio-gas plant establishment, IOT based establishment and some more. We can give a keen framework to gathering the milk from the people and paying them. Every rancher will be given a RFID card.

Whenever the individual goes to the milk community for providing the milk he have to swipe his RFID card and afterward he needs to put his milk can on the savvy gadget which will gauge the amount of milk. The amount alongside the card number will be put away in the cloud.

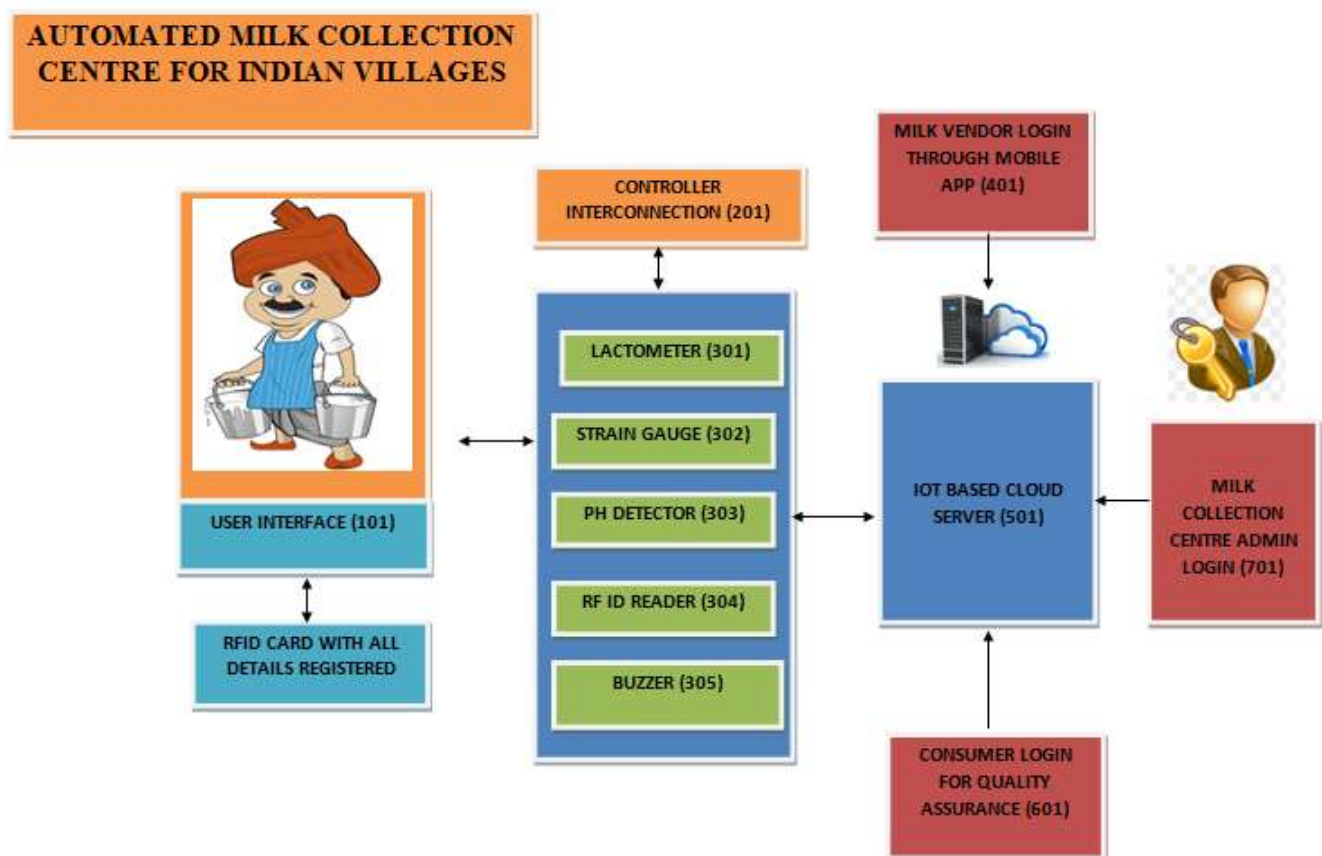


FIGURE 2: BLOCK DIAGRAM OF THE AUTOMATED MILK COLLECTION CENTRE FOR INDIAN VILLAGES

1. The Module 101 represents the user interface to the proposed system wherein the milk vendor carries a RFID tag that stores identify number of the particular vendor.
2. The module 201 represents the controller interface and its interconnection with lactometer (301), Strain Gauge (302), PH detector (303), RF ID Reader (304) and a buzzer (305).

3. The module 401 represents the milk vendor login through mobile app and its access to cloud server (501).
4. The module 601 represents the consumer login to access the cloud server to check the quality of the milk supplied for them daily.
5. The module 701 represents admin login to be maintained by the milk collection centre to organize the data.

A Village Level Milk Collection Center in-control is answerable for acquiring and taking care of huge volumes of milk from numerous individual milk producers. We can give a savvy framework to gathering the milk from the people and paying them. Each rancher will be given a RFID card. Whenever the individual goes to the milk community for providing the milk he have to swipe his RFID card and afterward he needs to put his milk can on the shrewd gadget which will gauge the amount of milk. The amount alongside the card number will be put away in the cloud. Toward the finish of consistently the bills will be created for each rancher so that in-control can pay them. Administrator will be given a versatile App utilizing which he can screen all boundaries. This framework has the capacity to decrease milk deterioration essentially and whenever utilized overall at that point would create in huge loads of information that will assist with improving milk transportation as well as consumer loyalty. A definitive objective of the undertaking is to profit the milk creation industry and the end shopper of the dairy items.

WE CLAIM

1. A system as claimed in claim 1, wherein the system is an automated milk collection centre for Indian villages.
2. A system as claimed in claim 2, wherein the system uses RFID reader to identify the milk vendor details fed in the server.

3. A system as claimed in claim 3, wherein the system estimates the quality of the milk provided by the vendor by using lactometer and PH sensor.
4. A system as claimed in claim 4, wherein the system estimates the quantity of the milk poured by the vendor by using load cell.
5. A system as claimed in claim 5, wherein the system provides an unique mobile application with individual login for vendor, customer and admin to access the portal.
6. A system as claimed in claim 6, wherein the system strongly ensures quantity and quality of the milk supplied by the vendor..

Dated this 13th NOVEMBER 2020

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AUTOMATED MILK COLLECTION CENTRE FOR INDIAN VILLAGES

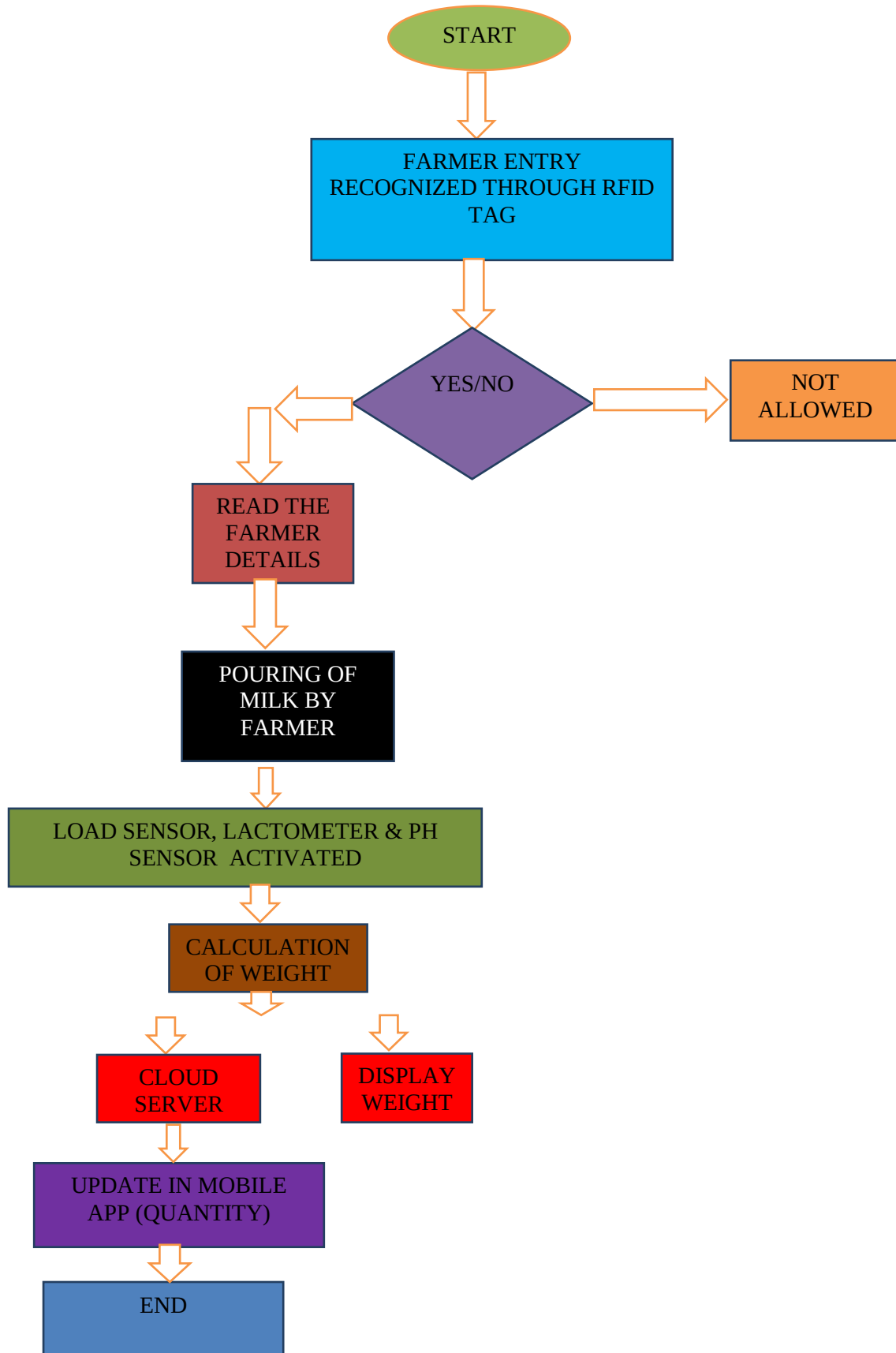


FIGURE 1: FLOW CHART OF THE AUTOMATED MILK COLLECTION CENTRE FOR INDIAN VILLAGES

AUTOMATED MILK COLLECTION CENTRE FOR INDIAN VILLAGES

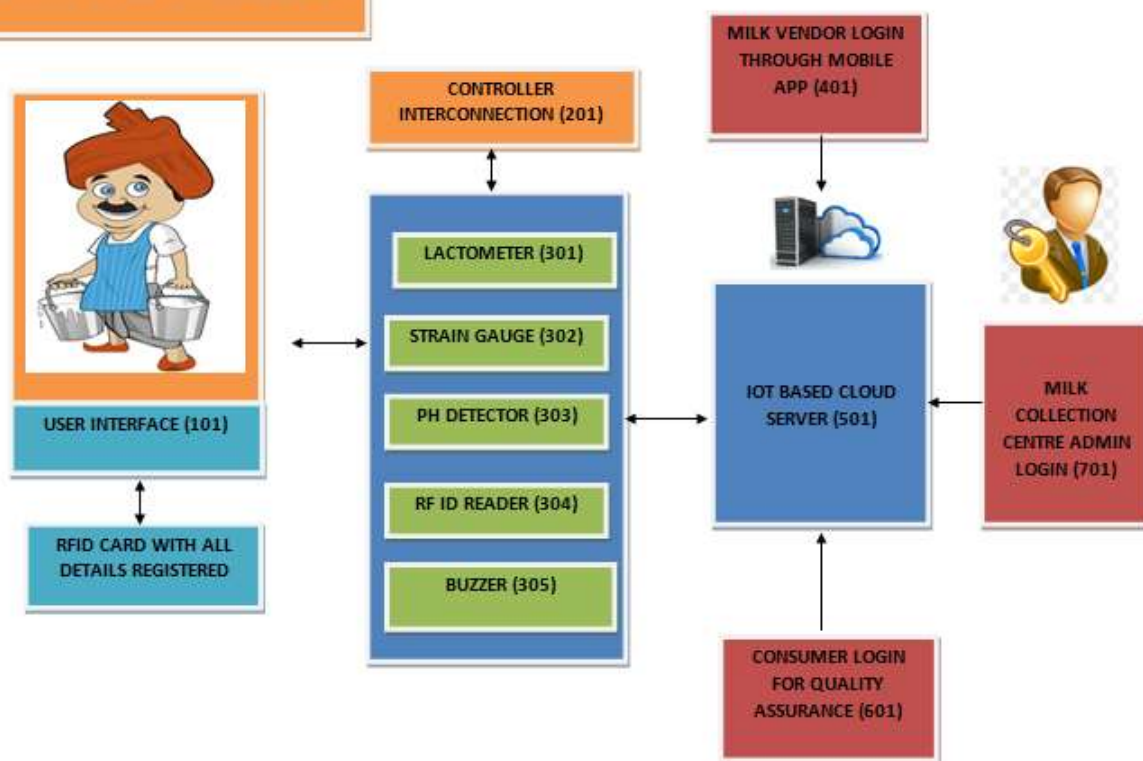


FIGURE 2: BLOCK DIAGRAM OF THE AUTOMATED MILK COLLECTION CENTRE FOR INDIAN VILLAGES

Dated this 13th NOVEMBER 2020

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AUTOMATED MILK COLLECTION CENTRE FOR INDIAN VILLAGES

WE CLAIM

1. A system as claimed in claim 1, wherein the system is an automated milk collection centre for Indian villages.
2. A system as claimed in claim 2, wherein the system uses RFID reader to identify the milk vendor details fed in the server.
3. A system as claimed in claim 3, wherein the system estimates the quality of the milk provided by the vendor by using lactometer and PH sensor.
4. A system as claimed in claim 4, wherein the system estimates the quantity of the milk poured by the vendor by using load cell.
5. A system as claimed in claim 5, wherein the system provides an unique mobile application with individual login for vendor, customer and admin to access the portal.
6. A system as claimed in claim 6, wherein the system strongly ensures quantity and quality of the milk supplied by the vendor..

Dated this 13th NOVEMBER 2020

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FORM 9

THE PATENT ACT, 1970
(39 of 1970)
&
THE PATENTS RULES, 2003

REQUEST FOR PUBLICATION

[See section 11A (2) rule 24A]

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Dated **13/11/2020 00:00:00** under section 11A(2) of the Act.

Dated this (Final Payment Date):-----

Signature

Name of the signatory

To,
The Controller of Patents,
The Patent Office,
At Chennai

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